A Simulation-Based Multi-Criteria Charging Station Recommendation Framework for Electric Vehicles Using SUMO and TraCI

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***Abstract*—The rapid growth of three wheeler electric vehicles – Easy Bikes, Electric-Rickshaws and Electric-Vans -all over the Bangladesh has great impact of unsupported charging demand on the urban power infrastructure. It isn't prepared for a peak classification. Lack of intelligent charging management systems reduces charging station utilization efficiency, increases traffic congestion and the risk of local power imbalance. To address those challenges, the paper represents a simulation-based Electric Vehicle charging station recommendation framework. It is developed using Eclipse SUMO microscopic traffic simulator, which is integrated with Python and TraCI. With different Electric Vehicle flows operating on multiple routes, a realistic urban mobility scenario was created. The scenario is supported by both-directional charging stations with realistic battery specifications. The suggested recommendation engine continuously observes -vehicles SoC, travel distance, slot-occupancy. Along with monitoring instant electricity load, the proposed framework applies multi valued decision model. With intelligent station selection enabled, recommendations are made periodically in an advisory decision-making mode to preserve traffic stability. Experimental results present that proposed framework gains a potential recommendation rate of 90.1%, which exceeds the nearest only baseline(66%). It also demonstrates effective congestion avoidance capabilities. The simulation recorded a peak charging demand of 1.37 MW with 40 simultaneously charging vehicles. This reflects realistic urban load stress scenarios. The resulting dataset supports machine learning based Electric Vehicle load forecasting and smart grid planning studies [8] within the emerging Electric Vehicle ecosystem.**

***Index Terms*—Electric Vehicles, Easy Bike, E-Rickshaw, Charging Recommendation, SUMO, TraCI, Smart Grid, Load Balancing, Bangladesh**

I. Introduction

The global adoption of electric vehicles (EVs) has accelerated as a strategy to prevent greenhouse gas (GHG) emissions from the transportation sector, which is responsible for approximately 38% of total global GHG production [1]. In Bangladesh, this change has taken an individual form: instead of personal passenger cars, battery-powered three-wheeled vehicles – locally known as Easy Bikes and E-Rickshaws – have become the main intercity transportation medium [9].

Khulna, the third largest city in Bangladesh and an important industrial hub, is a case in point. Thousands of electric three-wheelers ply the streets of Khulna every day, providing affordable, low-emission urban mobility and supporting the livelihoods of a significant driver population [2]. According to Khulna's traffic department, there are 30,000+ battery-driven easy bikes in the city, with Khulna City Corporation (KCC) having licensed 8,000 and processing approvals for another 2,000 [2].

In spite of these facilities, EV charging in Khulna remains almost completely imbalanced. Drivers select charging stations based only on proximity and familiarity, which results in intense demand at popular stations during morning and evening rush hours. Transformers serving to those overloaded nodes - experience thermal stress, voltage drops, and failures. This results repair costs and disruption of driver service [14]. In the meantime peripheral stations remain continuously unused, which highlights the infrastructure investment. Electric Vehicle load on the electricity grid is projected to reach 780 TWh by 2030 [3], [4] globally. It makes intelligent charging coordination an urgent operational priority [16].

Previous work on EV charging management has mainly addressed advanced-economic contexts where vehicles are freely routable [5], [6] and have the advantage of inter-vehicle navigation. There are 2 limitations in the context of Khulna city which are absent in this literature : (i) Easy Bikes and Electric Rickshaws and Electric Vans follow specific corridor routes, as SUMO - the simulator used enforces one-way lane assignment; (ii) charging infrastructure is scattered and heterogeneous in terms of capacity, power rating and geographical distribution [15]. This is a purpose-built simulation framework that realistically models the Khulna fleet and designs around it rather than ignoring routing limitations.

The paper represents a SUMO+TraCI simulation framework that models a heterogeneous fleet of continuous 100 vehicles(Easy Bikes, Electric Rickshaws, Electric Vans) on 8 road-routes [10], [11]. It monitors real time battery SOC and also issues per vehicle charging station recommendations through a multi criteria scoring models. This framework distinguishes between potential routing – when a directed network path exists. It also suggests only output when directional limitations prevent routing. In comparison with nearest only baseline, significant improvements are seen in potential recommendation rates, congestion avoidance and genius handling of high priority charging events. The contributions of this work are: (i) a calibrated SUMO simulation of Khulna city's heterogeneous Electric Vehicle fleet with physics-based battery modeling on 3 vehicle classes; (ii) a multi-criteria recommendation algorithm that clearly maintains SUMO directed routing; (iii) comparative evaluation with nearest only baseline using four performance metrics; (iv) a 33,721 recorded vehicle centric dataset.

II. Related Work

Planning of EV charging management research infrastructure, expands real time routing, timesteps. We study every thread to indicate our work position.

A. Infrastructure Planning and Site Selection

Wang et al. [2], [5] formulated charging station selection as a multi-objective optimization problem. It reduces travel time, grid pressure and cost through integer programming. Hussain et al. [6] applied analytical classification. It was Analytic Hierarchy Process(AHP). The methods combined with TOPSIS to rank candidate stations according to accessibility, demand density, and grid capacity. These methods inform long-term major decisions but do not indicate real-time dispatch across existing networks. Rahman et al. [1] surveyed adoption trends specific to Bangladesh, identifying coordination gaps, which are directly correlated to this study. The study has also been conducted on the socio-economic and environmental impacts of battery-powered auto-rickshaws in Khulna city in accordance of Bangladeshi urban areas [15]. It is estimated that autorickshaws consume about 4,660 megawatt hours of electricity per day in Bangladesh [13].

B. Real-Time Routing and AI-Based Selection

Current research explored reinforcement learning & AI based frameworks for dynamics Electric Vehicle charging distribution. These methods [7], [17]. target to balance the load at stations, to lessen traffic, decrease peak-demand through adaptive routing suggestion [12]. Microscopic traffic simulation platforms like-SUMO has been widely implemented for traffic optimization. More implementation for signal control, energy aware mobility modeling [10], [19]. GIS-based multi-criteria decision-making approach for locating EV charging stations has also been demonstrated [18] By the way fixed work aggregated real time suggestive recommendation of charging in to the heterogeneous EV fleet of South Asian cities.

C. Load Forecasting and Dataset Generation

Siddiqui et al. [8] suggested DeepBoost—a stacked ensemble of LSTM, XGBoost, CatBoost, and Linear Regression for day ahead EV charging station load forecasting. It improved the MAE by 9.4% and 32.7%, respectively, compared to individual CatBoost and XGBoost on the ACN dataset. Their work establishes - enriched feature engineering – station occupancy, depth of queue, SoC distribution, time of day. The work drives forecast accuracy. The dataset produced by our simulation accurately captures these features, making it directly usable as input for our charging station style models applied to the Khulna context [9] .

D. Research Gap

No previous work addresses: (i) simulation-based recommendation for corridor-limited Electric Vehicle fleet like Easy Bikes; (ii) a clear suggestive mode, which preserves simulation integrity; or (iii) empirical load-balancing evaluation on a heterogeneous three-vehicle-type fleet with different battery capacities in a South Asian urban environment. This paper fills three gaps.

III. System Architecture

Here the proposed structure consists of five components: the Road Network and SUMO Traffic Simulation Engine, Heterogeneous EV Fleet, Charging Station Infrastructure, the TraCI Communication Layer with the Python based Charging Recommendation Engine and the Data Logging Module.

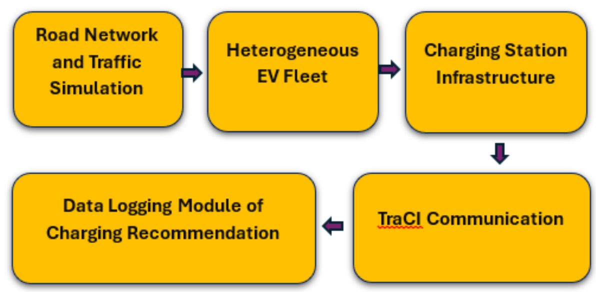


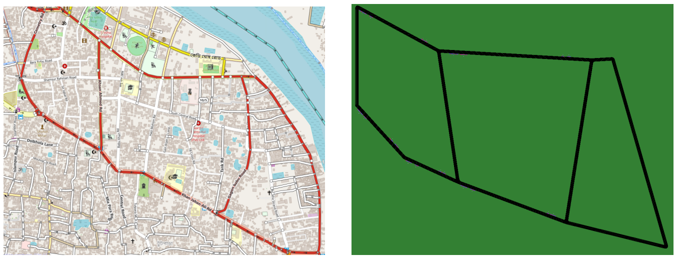
Fig. 1: System Architecture of the Proposed EV Charging Station Recommendation.

A. Road Network and Traffic Simulation

The simulated network creates a representative model of a part of Khulna city. It consists of 11 directed edges(E0-E10), bidirectional 8 designated routes(4 – forward, 4 - reverse). Traffic Light System controls the intersection model with realistic phase timing. The simulation runs from t=0 to t=600s with a 0.1s step resolution, with data collected every integer second. Time-to-teleport is disabled to prevent artificial vehicle movement or routing. Lateral resolution is set to 0.8m to properly model the narrow lanes of Khulna city, as shown in Fig. 2.

B. Heterogeneous EV Fleet

To model Khulna's real Electric Vehicle ecosystem, three vehicle classes with different battery configured: Easy Bikes(34 types), Electric Rickshaws(33 types), Electric Vans(33 types). Total 100 continuous flows across the network. From 2 to 81 vehicles per hour-launce vehicle through the 600 seconds simulation. Each vehicle has a SUMO battery device with specific values: 345 kg vehicle mass, 0.760 air drag coefficient, 0.0155 rolling drag, 81% propulsion efficiency, and 64% recovery efficiency [20]. Table I summarizes the three



(a) (b)

Fig. 2: (a) A particular region's map of Khulna city from Open Street Map. (b) Simulated road (custom) network of Khulna city in SUMO, consisting of 11 directed edges (E0–E10) with 8 bidirectional routes and traffic-light-controlled intersections (red dots indicate junction nodes).

vehicle fleet classes. Significantly, the easy bikes carry 12,000 Wh batteries, whereas the e-rickshaws and e-vans are equipped with smaller batteries with 5,760–7,200 Wh batteries. This scenario reflectes the actual mixed-fleet composition deployed in Khulna. During simulation vehicles enter with different initial SoC levels(12.9% - 43.2%), that reflects realistic field conditions.

TABLE I: Heterogeneous EV Fleet Configuration

Vehicle Type	**Count**	**Battery (Wh)**	**Max Speed (m/s)**
Easy Bike	34	12,000	8.33
E-Rickshaw	33	5,760–7,200	6.94
E-Van	33	5,760–7,200	5.56

C. Charging Station Infrastructure

10 charging stations are implanted as simulation parking areas over 2-way, 5-road corridors. Each of them has separate charging sub areas and waiting sub areas. The charging sub areas have SUMO charging device parameters like-charging power, efficiency of charging. They enable local SUMO battery rechargeable. Table II explains detailed configuration. Here the total system capacity is 62 charging slots and 49-waiting slots. Station power rating range is : 25-50 kW with 95% efficiency.

TABLE II: Charging Station Configuration

Station	**Chg. Slots**	**Wait. Slots**	**Power (kW)**	**Eff.**
pa_1 (fwd)	4	3	25	95%
pa_1_rev	4	3	25	95%
pa_3 (fwd)	5	4	30	95%
pa_3_rev	8	6	40	95%
pa_5 (fwd)	4	3	25	95%
pa_5_rev	4	3	25	95%
pa_7 (fwd)	6	5	35	95%
pa_7_rev	6	5	35	95%
pa_8 (fwd)	5	3	50	95%
pa_8_rev	10	6	50	95%
Total	**62**	**49**	25–50	95%

D. TraCI Communication Layer

TraCI gives a socket based API. It interacts with SUMO and Python. In every integer second, the controller gathers battery parameters (`device.battery.chargeLevel`, `device.battery.capacity`, `device.battery.totalEnergyConsumed`), vehicle location, speed, current road ID, lane ID and stop status. It uses `traci.simulation.findRoute()` to evaluate topological reachability. This is an important step for directional constraints—and `traci.lane.getShape()` to calculate Euclidean station distances. Where a route change is possible and the vehicle's SoC drops below the threshold value of 30% warning. The controller reroutes the vehicle to its assigned parking area(used as charging area here) by calling `traci.vehicle.setRoute()` and `traci.vehicle.setParkingAreaStop()`.

E. Data Logging Module

Each recommendation event is logged as 28 attribute CSV record:- timestep, vehicle-type, current edge, lane, speed, SoC, battery capacity, priority level and full metadata (ID, distance, power rating, charging/waiting availability, occupancy rate) for both the nearest, 2nd nearest stations, plus textual reasoning string of the recommended station. 1,169 recommendation events are generated in the simulation. Moreover a vehicle-centric dataset contains 33,721 timestep vehicle records with system level features. It includes total active vehicles, charging vehicles, average SoC, system power demand, and per-vehicle status and metrics.

IV. Charging Recommendation Model

A. Multi-Criteria Scoring Formulation

At simulation time t, for a vehicle v requiring charging, the proposed station-S^* reduces a composite score over the set of N=10 available stations:

$$S^* = \arg\min_i \left(w_1 D_i + w_2 Q_i + w_3 L_i\right) \tag{1}$$

Where, D_i = Euclidean distance from vehicle v to station s_i (m), Q_i = number of vehicles currently in the waiting queue at s_i, and L_i = instant charging power load at s_i (kW). Weighting factors w_1 = 0.5, w_2 = 0.3, w_3 = 0.2 were set to prioritise proximity while moderating congestion and load effects. Rankings are calculated using Euclidean distance for computational efficiency at every 1 second polling interval. The top two candidates (closest and second-closest). Then go through the decision logic below.

B. SoC Priority Classification

Based on the vehicle's current SoC , each recommendation event is tagged with one of 3 priority levels : CRITICAL (SoC $<$ 20%), HIGH (20% $\leq$ SoC $<$ 30%), and MEDIUM (30% $\leq$ SoC $<$ 50%). Priority-level tagging serves a significant purpose. It controls the urgency of console alerts during simulation (CRITICAL and HIGH events trigger verbose output).

- **CRITICAL** (SoC $< 20\%$): immediate action required — risk of battery depletion in transit.
- **HIGH** ($20\% \leq$ SoC $< 30\%$): urgent — vehicle should charge at the next available opportunity.
- **MEDIUM** ($30\% \leq$ SoC $< 50\%$): proactive recommendation — vehicle can reach the station comfortably.

C. Directional Constraint and Advisory Mode

A notable innovation of this framework is SUMO's explicit handling of directional routing constraints. In Khulna, easy bikes and e-rickshaws operate on specific corridor routes. SUMO enforces one-way lane assignments and cannot teleport vehicles to the opposite end. As a result, the framework operates in two modes:

Rerouting Mode: When, traci.simulation.findRoute() - returns a valid directed path from the vehicle's current end to the target station end, with a positive route length, the framework calls `setRoute()` and `setParkingAreaStop()` to redirect the vehicle. A return route to the vehicle's original destination is added if possible.

Advisory Mode: When no valid guided path exists, because of the target station is at an unreachable distance from the vehicle's recent running-structure, the framework logs the best scoring station as suggested recommendation. It preserves simulation integrity and produces useful data for real-world notification systems.

D. Four-Rule Decision Logic

Given the nearest station s_1 and second nearest station s_2. The recommendation algorithm applies the following rules sequentially:

1) **Rule 1**: If s_1 has a free charging slot $\rightarrow$ recommend s_1 (Rerouting Mode if path exists).
2) **Rule 2**: If s_1 is full for charging but s_2 has a free charging slot $\rightarrow$ recommend s_2.
3) **Rule 3**: If s_2 also lacks charging slots but s_2 has waiting capacity $\rightarrow$ recommend s_2 (waiting queue).
4) **Rule 4**: If both s_1 and s_2 are fully saturated $\rightarrow$ recommend s_1 (Advisory Mode, flag for retry).

A 30 second cooldown per vehicle prevents recommendation changes. Queue management uses FIFO propagation:- when a charging vehicle leaves, the next waiting vehicle will automatically promoted to a charging slot via `traci.vehicle.resume()` and new `setParkingAreaStop()` call. That maintains orderly slot throughput without outside interference.

V. Simulation Setup and Parameters

Table III summarizes the complete simulation parameter set. `Test1.sumocfg` (SUMO configuration) specifies the road network (`Test1_with_TLS.net.xml`), vehicle flows (`Test1.rou.xml`) and charging infrastructure (`Test1.chargingstations.add.xml`). Battery output, charging station output, trip information and summary statistics are collected via SUMO's native output facility. The Python controller (`charging_recommendation`) connects through TraCI, iterates in 1-second integer steps, and exports recommendations datasets to timestamped CSV files after simulation completion at t = 600 seconds.

TABLE III: Key Simulation Parameters

Parameter	Value
Simulator	SUMO (Eclipse)
Interface	TraCI (Python 3.x)
Simulation duration	600 s
Time step resolution	0.1 s
Number of vehicle flows	100
Vehicle classes	Easy Bike, E-Rickshaw, E-Van
Number of routes	8 (4 fwd + 4 rev)
Network edges	11 directed
Charging stations	10 bidirectional
Total charging slots	62
Total waiting slots	49
Charging power range	25–50 kW
Charging efficiency	95% [21]
SoC trigger threshold	50%
CRITICAL priority	SoC $< 20\%$
HIGH priority	$20\% \leq$ SoC $< 30\%$
Recommendation cooldown	30 s
Scoring weights (w_1, w_2, w_3)	0.5, 0.3, 0.2

VI. Results and Discussion

A. System Load and Charging Statistics

The simulation ended at 600 s. It generates 33,721 vehicle centric records(one vehicle per second) and 1,169 charging recommendation events. The system recorded a peak instant charging power of 1,370 kW with 40 vehicles charging concurrently. That reflects near saturation of the 62 available charging areas during peak demand time. Across the simulation the average charging power was 963.7 kW, the total energy consumption for charging was 127.5 kWh. These load metrics highlights that there is an important charging demand on the network. It makes intelligent station selection essential to prevent local transformer overload.

B. Recommendation Volume and Priority Distribution

Table IV shows the priority level breakdown of the 1,169 recommendation events. HIGH-priority events (SoC 20%–30%) dominated at 42.7%, followed by MEDIUM (34.6%) and CRITICAL (22.7%). The average SoC at the time of recommendation was 27.7% (σ = 8.5%), with a minimum of 12.0% and a maximum of 46.5%. It ensures that the 50% trigger activates recommendations with sufficient battery margin. The average Euclidean distance to the nearest station at the time of recommendation was 84.3 m (σ = 62.0 m, min = 0.5 m, max = 256.6 m). It is indicating adequate station density for the modelled fleet.

C. Recommendation Outcome Analysis

Table V shows the distribution of the results of recommendations. The main result — when charging slots were

TABLE IV: Recommendation Priority Distribution

Priority	SoC Range	Events	Pct.
CRITICAL	<20%	265	22.7%
HIGH	20%–30%	499	42.7%
MEDIUM	30%–50%	405	34.6%
Total	—	**1,169**	**100%**

available, 59.0% (689 events)—was routing to the nearest station. In 24.1% of cases (282 events), the nearest station was fully occupied and vehicles were successfully directed to the second nearest station with available capacity. It is demonstrating genuine load diversion. Where both stations were saturated, Waiting slot overflow management handled 7.7% (90 events), while 9.2% (108 events) triggered Advisory Mode.

TABLE V: Recommendation Outcome Analysis

Outcome	Events	Pct.
Nearest station (slot available)	689	59.0%
2nd nearest (nearest full)	282	24.1%
Waiting slot overflow	90	7.7%
Advisory Mode (both saturated)	108	9.2%
Total	**1,169**	**100%**

D. Station Utilisation Distribution

Station pa_8 received the highest recommendation share (32.4% of all events). Which reflects its large slot capacity (5 charging + 3 waiting forward, 10 + 6 reverse) and power rating up to 50 kW combined with a central network position. Stations pa_7 and pa_3_rev came in second and third with 14.7% and 14.0% respectively. Significantly, all ten stations received at least 1.5% of recommendations, confirming that the multi-criteria scoring model [7], [12] widely distributes demand.

E. Comparative Evaluation: Proposed vs. Nearest-Only Baseline

To measure the benefit of the structure, we define a nearest-only baseline that always recommends the Euclidean nearest station without considering occupancy, queue, or load. Table VI shows 4-performance metrics computed from the 1,169-event dataset.

TABLE VI: Performance Comparison: Proposed vs. Nearest-Only Baseline

Metric	Proposed	Baseline
Feasible Rec. Rate (FRR)	90.1%	66.0%
Congestion Avoid. Rate (CAR)	37.1%	0.0%
Hi-Priority Smart Dec. (HPSDR)	27.9%	0.0%
Avg. Distance (m)	100.1	84.3

The following subsections explain each metric in depth - its definition, calculation method, numerical values and significance in this work.

1) Feasible Recommendation Rate (FRR): FRR is the fraction of all recommendation events , where at least one free charging slot was available at the station recommended by the algorithm, at the time of the recommendation. A recommendation is "feasible", if a vehicle is sent to the station can start charging immediately without waiting for a slot to become free.

Calculation Formula:

$$\mathrm{FRR} = \frac{N_{\text{rec. station}\geq 1 \text{ free slot}}}{N_{\text{total events}}} \times 100\% \tag{2}$$

For the proposed framework, the feasible events are those resolved by Rule 1 (689 events, nearest with free slot) plus Rule 2 (282 events, 2nd nearest with free slot) = 971 feasible events out of 1,169 total.

$$\mathrm{FRR\ (Proposed)} = \frac{971}{1{,}169} \times 100\% \approx 90.1\% \tag{3}$$

For the baseline, the nearest station is only possible int the 771 events (66.0% of 1,169) where there was a free slot at the nearest station , regardless of diversion. The baseline doesn't attempt to check or redirect to a better option.

Interpretation: An FRR of 90.1% means that 90 out of every 100 recommendation events, are taken to a station where it can immediately plug in. The 24.0 percentage-point improvement over the baseline quantifies the concrete benefit of occupancy-aware selection: vehicles are far less likely to arrive at a full station, waste time waiting, or drain their already-depleted battery in further searching.

2) Congestion Avoidance Rate (CAR): CAR computes the proportion of all recommendation events , where the proposed framework successfully diverted a vehicle away from a congested (fully occupied) nearest station to an alternative station with available capacity. It directly quantifies the load-balancing effectiveness of the recommendation engine.

Calculation Formula:

$$\mathrm{CAR} = \frac{N_{\text{nearest full \& alt. with free slots}}}{N_{\text{total events}}} \times 100\% \tag{4}$$

This corresponds exactly to Rule 2 events: 282 cases in which the nearest station was fully occupied but the second-nearest had capacity. The baseline, which blindly recommends the nearest station, contributes 0 such diversions.

$$\mathrm{CAR\ (Proposed)} = \frac{282}{1{,}169} \times 100\% = 37.1\% \tag{5}$$

$$\mathrm{CAR\ (Baseline)} = 0.0\% \tag{6}$$

Interpretation: A CAR of 37.1% means that more than one in three charging events, which would have overloaded a popular station are instead redistributed to a less congested alternative. In grid terms. This translates directly to reduced transformer peak loading, lower risk of thermal stress, and more uniform utilisation of the installed charging infrastructure. The baseline's CAR = 0% confirms that proximity-only dispatching cannot achieve any congestion relief by design.

3) High-Priority Smart Decision Rate (HPSDR): HPSDR measures the proportion of CRITICAL (SoC $<$ 20%) and HIGH (20% $\leq$ SoC $<$ 30%) priority recommendation events — those carrying the highest battery urgency — in which the framework successfully moved the vehicle to the second-nearest station because the nearest station lacked available charging slots. It captures intelligent decision-making precisely when the stakes are highest: when battery depletion in transit is a realistic risk.

Calculation Formula:

$$\text{HPSDR} = \frac{N_{\text{CRIT+HIGH: chose 2nd over congested 1st}}}{N_{\text{total events}}} \times 100\% \quad (7)$$

CRITICAL events: 265; HIGH events: 499; combined = 764 high-urgency events out of 1,169 total (65.4%). Of these, the framework successfully diverted to the second-nearest station (Rule 2 applied during CRITICAL/HIGH events) in 27.9% of all events = 326 events.

$$\text{HPSDR (Proposed)} = \frac{326}{1{,}169} \times 100\% = 27.9\% \quad (8)$$

$$\text{HPSDR (Baseline)} = 0.0\% \quad (9)$$

Interpretation: An HPSDR of 27.9% demonstrates that the framework doesn't apply a blanket nearest-station rule even under battery urgency — it actively reasons about availability and routes critically depleted vehicles to the best accessible station. The 27.9 percentage-point advantage over the baseline is particularly significant in the Khulna context, where running out of charge mid-route leaves an Easy Bike or E-Rickshaw stranded, blocking traffic and causing immediate income loss for the driver. The baseline achieves HPSDR = 0% because it has no mechanism to deviate from the nearest station regardless of urgency level.

4) Average Distance to Recommended Station: : This metric reports the mean Euclidean straight-line distance (in metres) from a vehicle's position at the moment of recommendation to the station it is recommended to visit. It serves as the primary cost metric to assess the trade-off incurred by congestion-aware routing: redirecting vehicles to alternative stations necessarily increases travel distance.

Calculation Formula:

$$\bar{d} = \frac{1}{N} \sum_{i=1}^{N} d_{\text{Eucl.}}(\text{pos}_i,\ s_{i,\text{rec}}) \quad (10)$$

where N = 1,169 total recommendation events, pos_i is the vehicle's geographic coordinates at event i and $s_{i,\text{rec}}$ is the centre coordinates of the recommended station. The Euclidean distance is measured using `traci.lane.getShape()` coordinate data polled at each integer second.

$$\bar{d}_{\text{Baseline}} = 84.3 \text{ m} \quad \text{(always the nearest station by definition)} \quad (11)$$

$$\bar{d}_{\text{Proposed}} = 100.1 \text{ m} \quad \text{(can be the 2nd nearest)} \quad (12)$$

Interpretation: The suggested framework asks vehicles to travel, on average, 15.8 m further than the nearest station. This is an accepted cost to avoide crowded stations. To put this tranasaction in to context, consider that the average SoC at recommendation time is 27.7% and the minimum battery capacity is 5,760 Wh. Even at an average propulsion power of $\sim$1.5 kW and a speed of 10 m/s. The additional 15.8 m adds less than 2 seconds of travel time and consumes a negligible energy increment well within the available battery margin. The trade-off is therefore, operationally inconsequential while the gains in FRR (+24.0 pp), CAR (+37.1 pp), and HPSDR (+27.9 pp) are substantial.

5) Summary Comparison and Interpretation: The table VI combines all four metrics. The proposed multi-criteria framework outperforms the closest only baseline on every effectiveness dimension while enduring a small distance penalty. The combined information is as follows :

FRR (90.1% vs 66.0%): 9 in 10 recommended stations have free slots immediately versus only 2 in 3 under naive routing. This reduces wasteful station to station re-querying by vehicles already at low SoC.

CAR (37.1% vs 0.0%): The framework actively load-balances by redirecting more than a third of all events away from congested hubs to underutilised alternatives, reducing transformer peak stress and prolonging infrastructure lifetime.

HPSDR (27.9% vs 0.0%): Even the most critically battery-depleted vehicles receive improved routing — the framework applies its highest priority triage precisely, while failures are most costly in the real world.

Avg. Distance (+15.8 m): The marginal additional travel introduced by smarter routing is 15.8 m — less than 2 vehicle lengths — and is operationally negligible relative to the energy and time benefits.

F. Limitations and Future Work

A simple network is included in the 600-second simulation window. Full scale deployment requires GPS trace calibrated OD matrices and different charging profiles from field surveys. Advisory Mode (9.2% of occurrences) can be reduced by increasing station density or adding a 3rd nearest candidate. Dynamic electricity, pricing and renewable energy at station nodes will be integrated in future work.

VII. Implications for Bangladesh

The 37.1% Congestion Avoidance Rate suggests that approximately one in three charging events that would otherwise overload a popular station can be redistributed. This can decrease the peak transformer loading and increase equipment lifetime. The resulting datasets (33,721 vehicle-centric + 1,169 recommendation events) accurately gives the feature space separated valued columns.

The per station recommendation distribution reveals that pa_8 and pa_7 corridors absorb inconsistent demand- which directly actionable information for capacity increasing. The open SUMO+Python architecture enables urban or city planners to test alternative station placements by modifying the XML configuration without specialised optimization skills.

VIII. Conclusion

This paper shows a SUMO+TraCI simulation-based charging station recommendation framework for heterogeneous EV fleet in Khulna city. A 100 flow simulation through 3- vehicle classes with varying battery capacities (12,000 Wh for Easy Bikes; 5,760–7,200 Wh for E-Rickshaws and E-Vans), eight routes and ten bidirectional stations generated 1,169 recommendation events over 600 seconds. Also it depicts the record of peak charging power of 1,370 kW with 40 vehicles simultaneously charging and consuming 127.5 kWh total energy.

The main contributions are: (1) a physics-calibrated SUMO fleet model with different battery configurations across three vehicle classes; (2) a multi-criteria recommendation algorithm with Advisory duality; (3) comparative evaluation demonstrating +24.0 pp potential Recommendation Rate, +37.1 pp Congestion Avoidance Rate, and +27.9 pp High-Priority Smart Decision Rate versus a nearest-only baseline; and (4) two complementary datasets (33,721 vehicle-centric records + 1,169 recommendation events).

Future work will be expanded to include GPS trace calibration of the full road graph of Khulna, dynamic pricing, model renewable energy at stations and evaluate demand-response schedules consistent with Bangladesh's National Energy Policy.

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